



TEMPERATURE

Changes in Temperature and Air pollutants across Atlantic provinces

5.60

Avg Monthly Temp

6.80

Avg Monthly Prediction

0.31

Trend(Deg C /Decade)

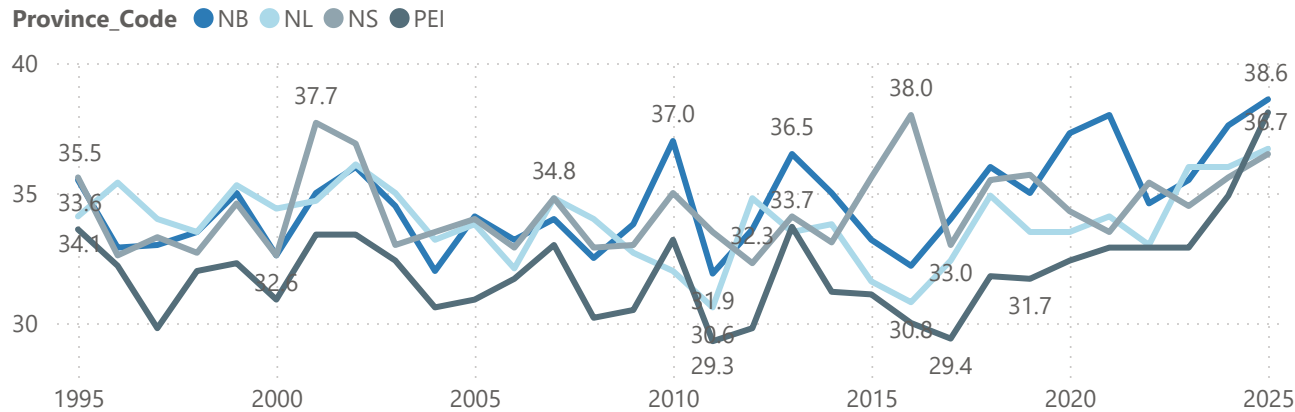
Good

Avg AQI

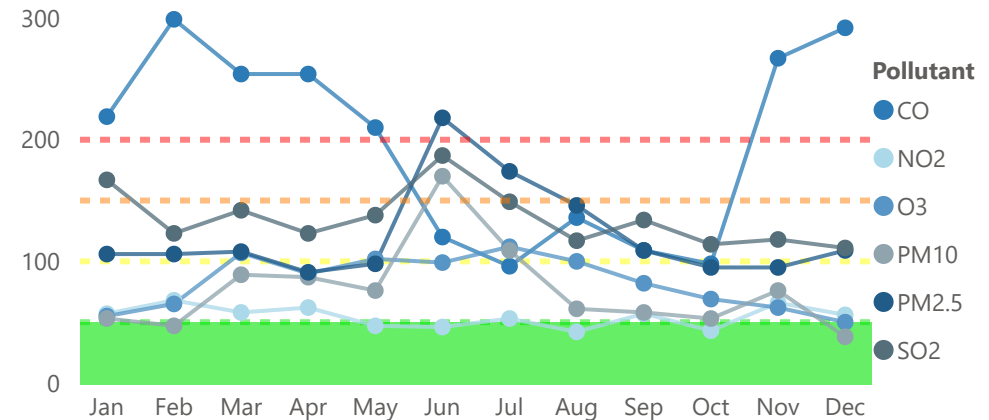
166

Unhealthy Days

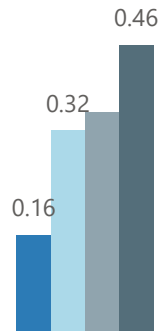
Max Temp (°C)



Worst Pollutant AQI



Trend (°C / decade) GHG Temp Correlation

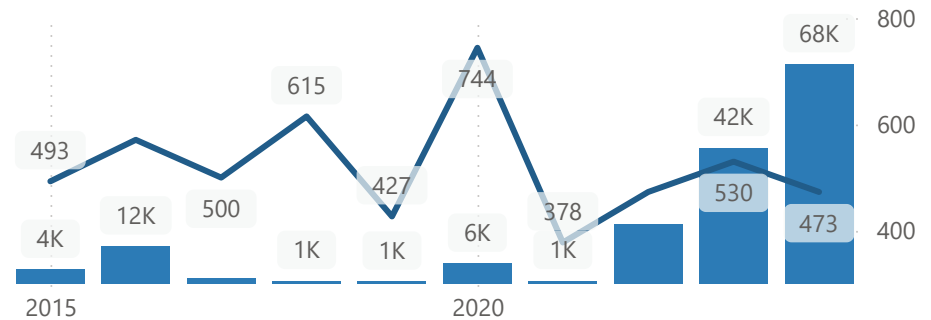


PEI

Correlation

Pollutant	Mean Temp (°C)
PM2.5	0.16
O3	0.15
PM10	0.08
CO	0.04
SO2	-0.01
NO2	-0.01

● Total Area Burned ● Total Fires



Is Atlantic, P...

^ ATLANTIC

■ NB

■ NL

■ NS

■ PEI

∨ ☐ REST OF CA...

Year

☐ 1995

☐ 1996

☐ 1997

☐ 1998

☐ 1999

☐ 2000

☐ 2001

☐ 2002

☐ 2003

☐ 2004

Pollutant

CO

NO2

O3

PM10

PM2.5

SO2

5.60
Avg Monthly Temp

-0.08
Temp Monthly
Anomaly (°C)

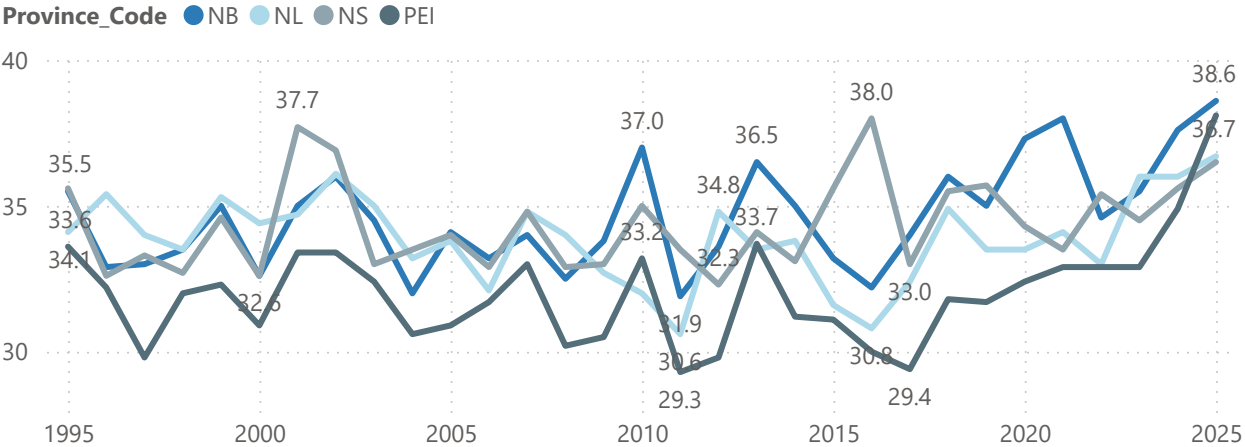
0.31
Trend(Deg C /Decade)

76.77
Avg Monthly Heatwaves

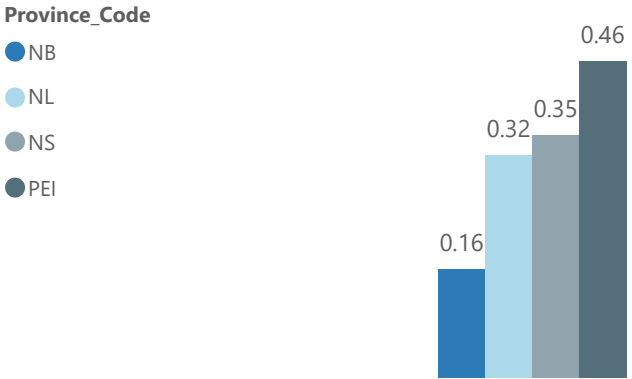
6.80
Average Prediction

- Provin...
☐ NB
☐ NL
☐ NS
☐ PEI
- Season
☐ Fall
☐ Spring
☐ Summer
☐ Winter

Max Temp (°C)

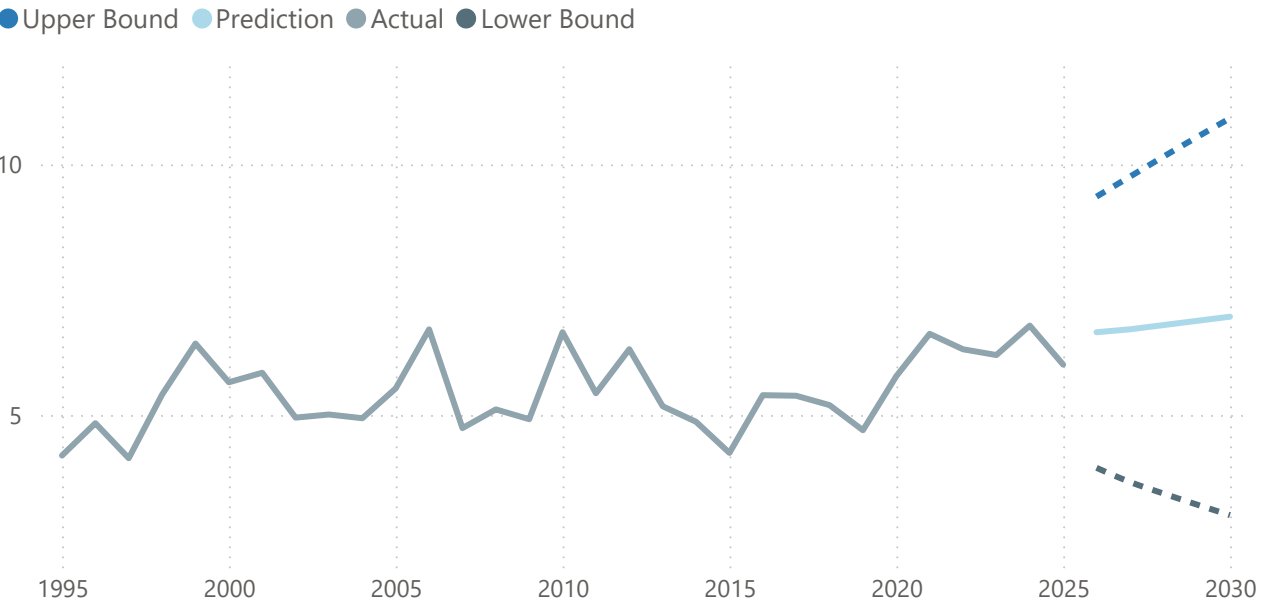
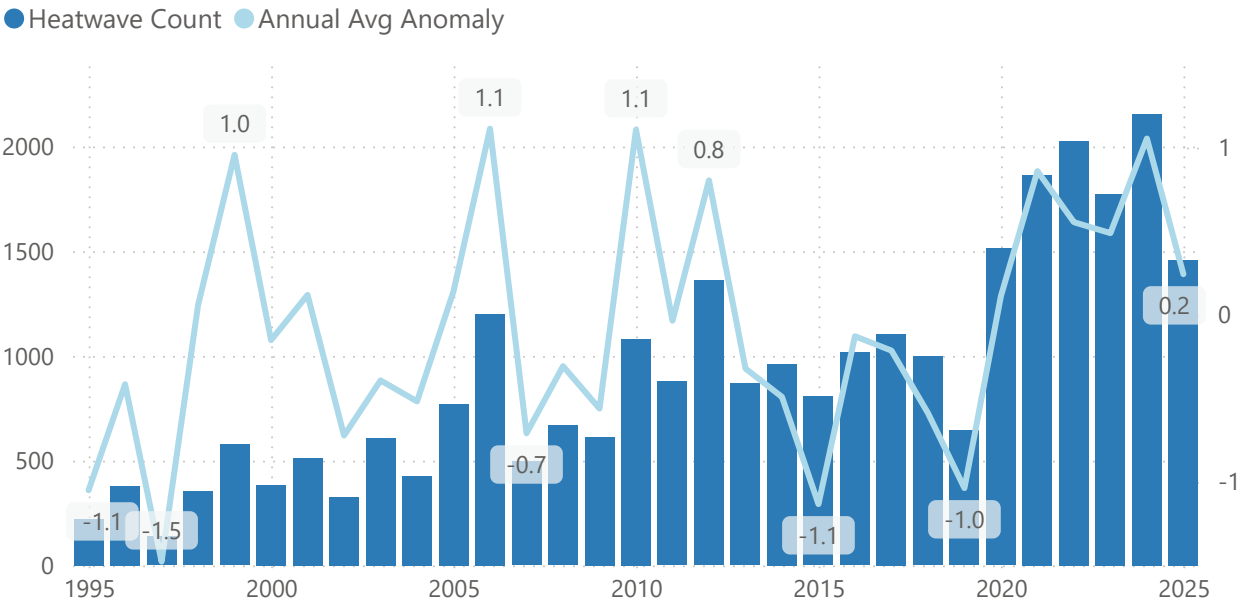


Trend (°C / decade)



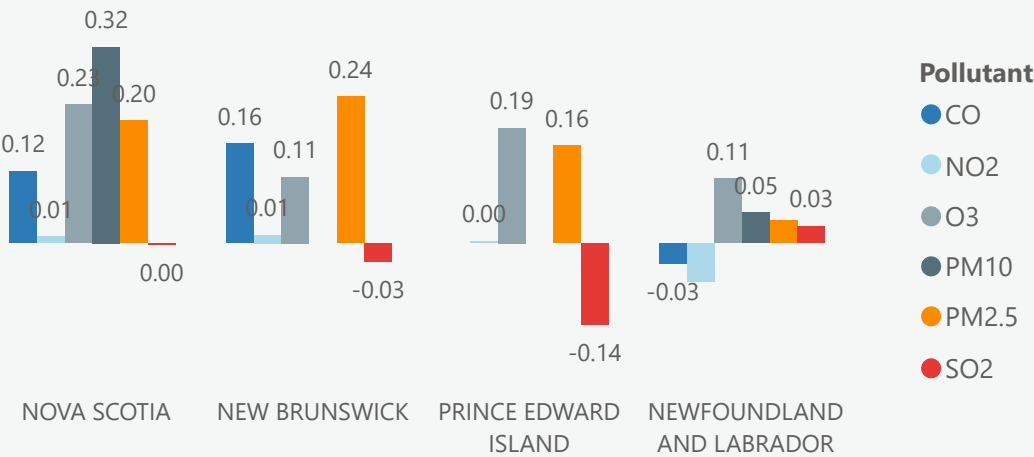
Model Comparison

OLS: MAE: 7.4763
RMSE: 8.3924
ARIMA: MAE: 1.5652
RMSE: 1.9148
SARIMA: MAE: 0.9765
RMSE: 1.2408

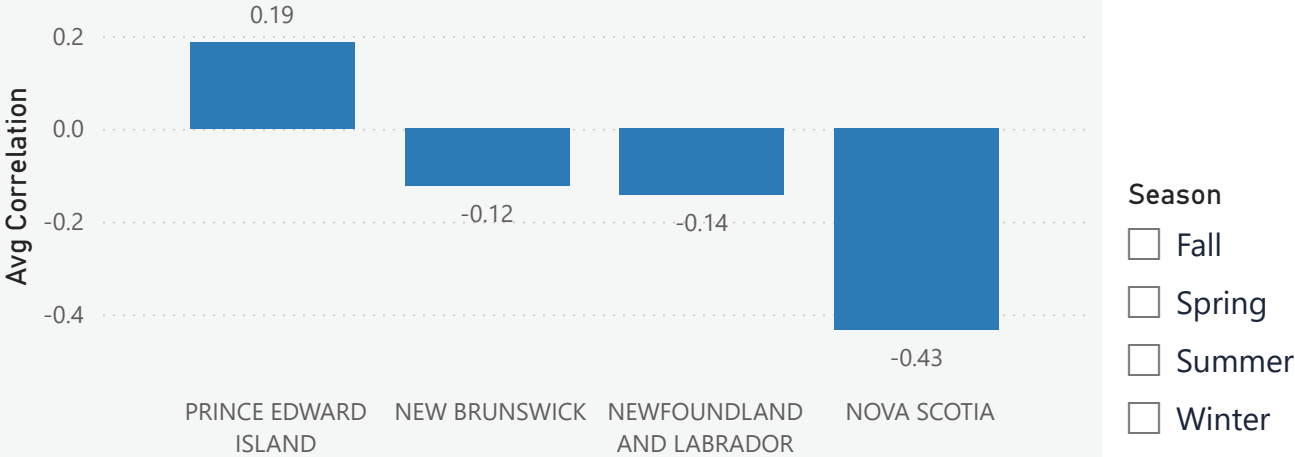


CORRELATION OF POLLUTANT CONCENTRATIONS & GREEN HOUSE GAS EMISSIONS WITH TEMPERATURE

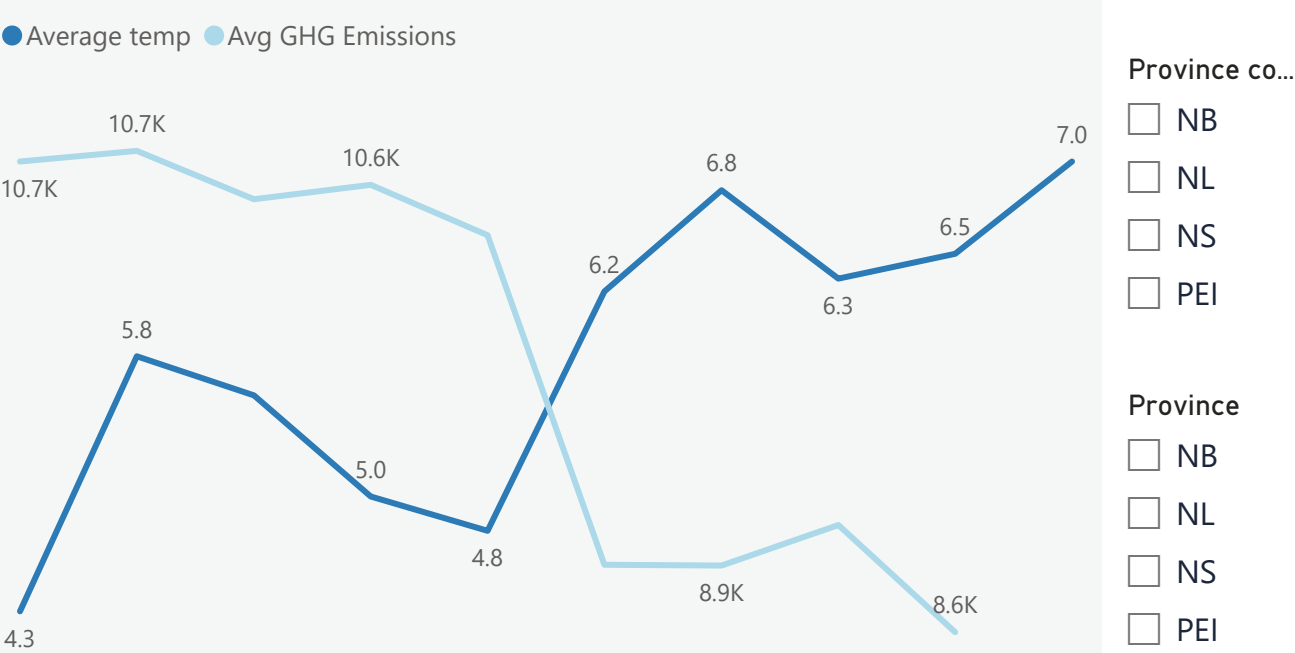
Pollutant Temperature Correlation



GHG Temperature Correlation



Pollutant	Mean Temp (°C)	Spd of Max Gust (km/h)	Total Rain (mm)	Total Snow (cm)
CO	0.04	-0.20	-0.08	-0.01
NO2	-0.01	-0.19	-0.03	0.00
O3	0.15	0.19	0.05	-0.01
PM10	0.08	0.05	-0.11	-0.09
PM2.5	0.16	-0.04	-0.12	-0.07
SO2	-0.01	-0.03	-0.07	-0.02





Good

Station AQI

34.7

Avg AQI -Station

O3

Dominant Pollutant

PM2.5

Second Dominant

0.16%

% Unhealthy Days

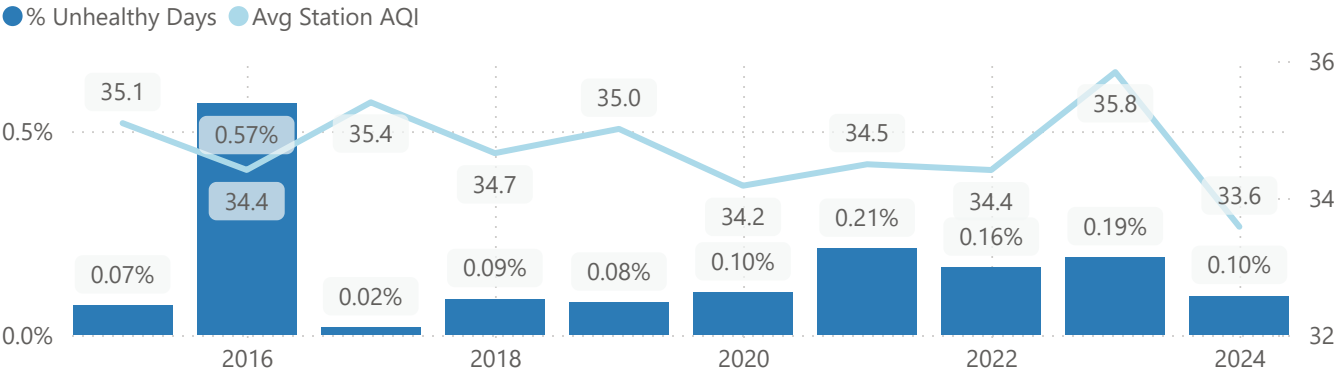
299

Worst Pollutant AQI

Overall AQI is GOOD ✓ (Avg: 34.7), with 0.2% unhealthy days. Trend is improving ↗, however watch for worst AQI level which reached 299 on 25 Feb 2016.

AQI Insight

% Unhealthy Days and Avg Station AQI



Province	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
NEW BRUNSWICK										
CO	2.8	2.5	2.5	2.4	2.4	2.2	2.4	2.2	2.5	2.3
NO2	9.7	8.4	9.1	8.7	8.6	6.9	6.7	7.3	7.2	6.5
O3	29.6	28.3	31.1	31.2	31.2	30.6	30.4	30.6	30.8	30.2
PM2.5	32.2	27.6	30.3	27.4	28.8	28.1	28.7	28.3	29.8	23.7
SO2	8.6	7.5	5.9	8.8	9.6	10.0	10.2	9.9	6.8	7.7
NEWFOUNDLAND AND LABRADOR										
CO	2.1	13.6	2.2	2.0	1.9	1.9	1.9	1.8	2.0	1.9
NO2	6.2	7.1	7.1	6.5	5.9	5.7	5.9	5.6	6.0	5.7
O3	31.4	31.0	31.1	29.7	29.6	29.9	30.2	29.9	30.2	30.5

Avg Station AQI



Clear all slicers

Is Atlantic, Provin...

ATLANTIC

NB

NL

NS

PEI

Year

2014

2015

2016

2017

2018

2019

2020

2021

2022

2023

2024

Season

Fall

Spring

Summer

Winter

Station_Type

AQI Station

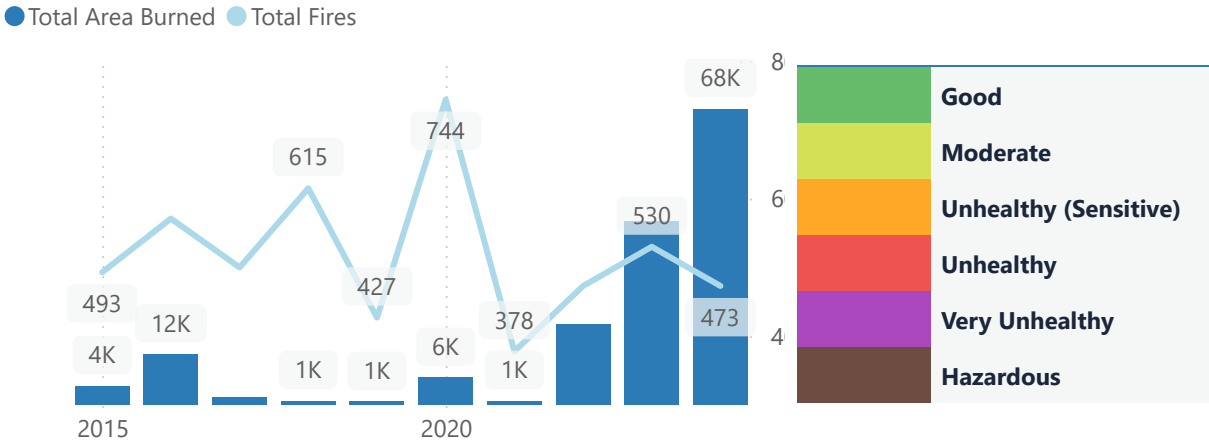
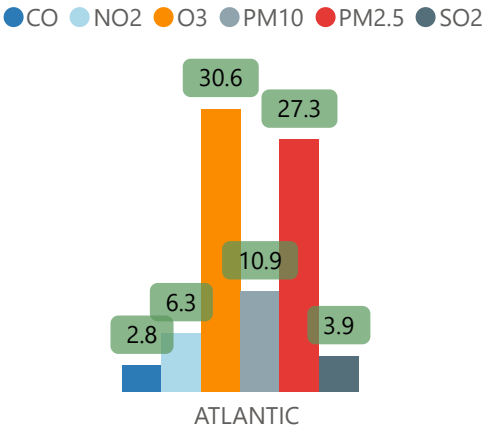
166

Unhealthy Days

CO, O3, SO2, PM2.5, PM10

Unhealthy Pollutants (>100 AQI)

Avg Pollutant AQI



↶

Is Atlantic, Provin...

▲

ATLANTIC

NB

NL

NS

PFI

Year

2015

2016

2017

2018

2019

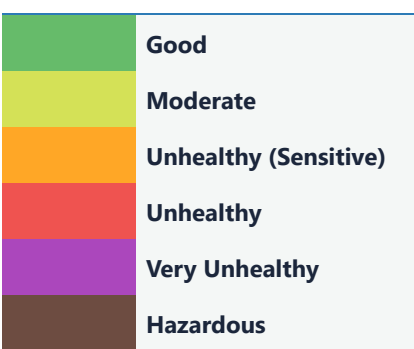
2020

2021

2022

2023

2024



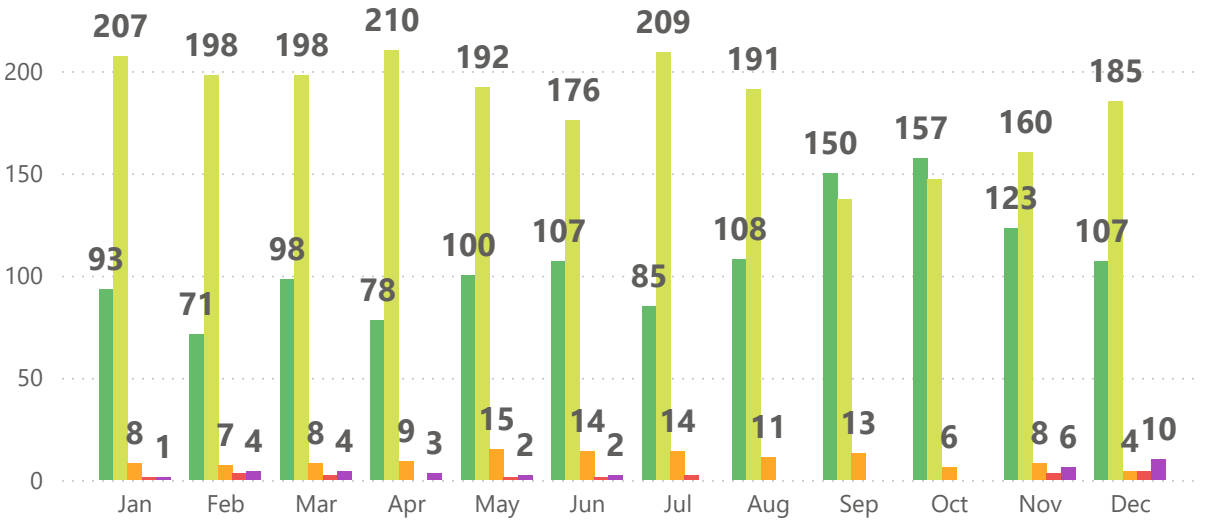
Pollutant

- CO
- NO2
- O3
- PM10
- PM2.5
- SO2

Worst Day: Pollutant AQI

Province_Code	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
[-] NL												
[+] CO	219	299	254	254	210	120	96	136	109	98	267	292
[+] PM2.5	80	67	66	71	71	218	174	146	90	95	74	109
[+] PM10	53	47	89	87	76	170	109	61	58	37	76	38
[+] O3	55	65	107	80	102	99	50	61	47	64	62	50
[+] SO2	54	42	84	72	45	73	21	28	40	107	60	62
[+] NO2	57	42	58	62	47	46	53	32	57	33	66	45
[-] NB												
[+] SO2	167	123	142	123	138	187	149	117	134	114	118	111
[+] PM2.5	106	106	108	71	74	120	94	91	109	83	95	98
[+] O3	40	50	76	84	85	88	67	61	40	55	46	42

Days in AQI Category



Station_Type

A.

C.



Good

Station AQI

38.5

Avg AQI -Station

O3

Dominant Pollutant

PM2.5

Second Dominant

1.19%

% Unhealthy Days

300

Worst Pollutant AQI

Overall AQI is GOOD ✓ (Avg: 38.5), with 1.2% unhealthy days. Trend is improving ↗, however watch for worst AQI level which reached 300 on 03 Sep 2023.

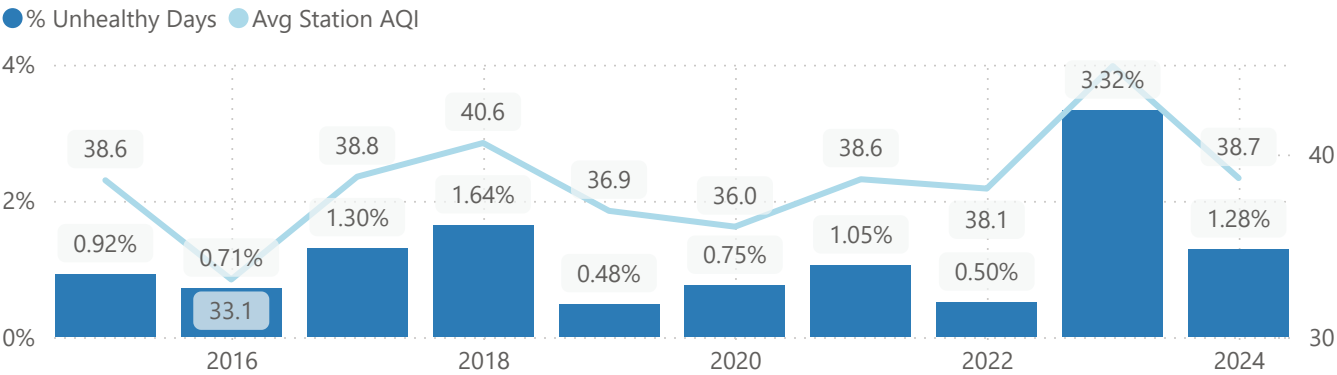
AQI Insight

Is Atlantic, Provin...

✓ ☐ ATLANTIC

✓ ☒ REST OF C...

% Unhealthy Days and Avg Station AQI



Avg Station AQI



Year

☐ 2015

☐ 2016

☐ 2017

☐ 2018

☐ 2019

☐ 2020

☐ 2021

☐ 2022

☐ 2023

☐ 2024

Season

☐ Fall

☐ Spring

☐ Summer

☐ Winter

Station_Type

☒ AQI Station

Province	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
ALBERTA	18.0	15.5	18.8	20.3	18.0	17.1	18.6	18.0	21.7	19.4
BRITISH COLUMBIA	18.8	15.5	19.7	19.8	18.1	17.7	18.3	18.3	18.6	16.8
MANITOBA	18.8	17.9	21.2	22.5	17.7	18.8	22.3	17.3	21.3	20.8
NORTHWEST TERRITORIES	14.3	11.0	14.0	13.7	14.5	12.5	14.5	15.6	19.7	19.5
NUNAVUT	25.9	20.1	20.7	20.2	21.2	29.0	29.7	27.9	30.1	29.2
ONTARIO	27.3	25.6	25.0	25.4	24.6	23.8	25.1	24.5	25.5	21.5
QUEBEC	26.0	22.2	25.2	25.2	24.2	23.4	24.4	24.5	21.9	24.0
SASKATCHEWAN	18.5	14.7	19.4	20.8	19.4	18.5	21.1	19.6	23.4	21.2
YUKON	23.3	20.8	22.0	20.6	24.6	23.7	24.6	22.8	21.9	22.1



2K

Unhealthy Days

CO, NO2, O3, SO2, PM2.5, PM10

Unhealthy Pollutants (>100 AQI)

↶

Year

2015

2016

2017

2018

2019

2020

2021

2022

2023

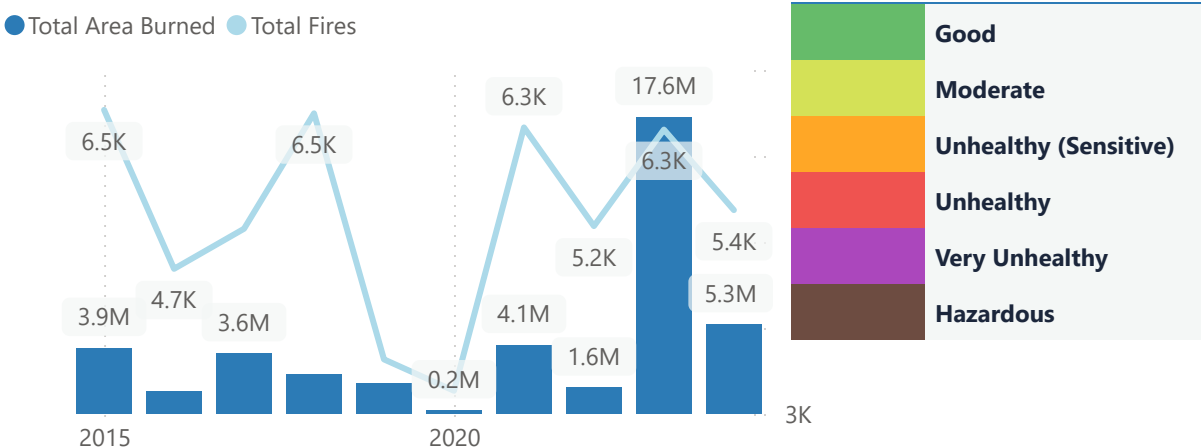
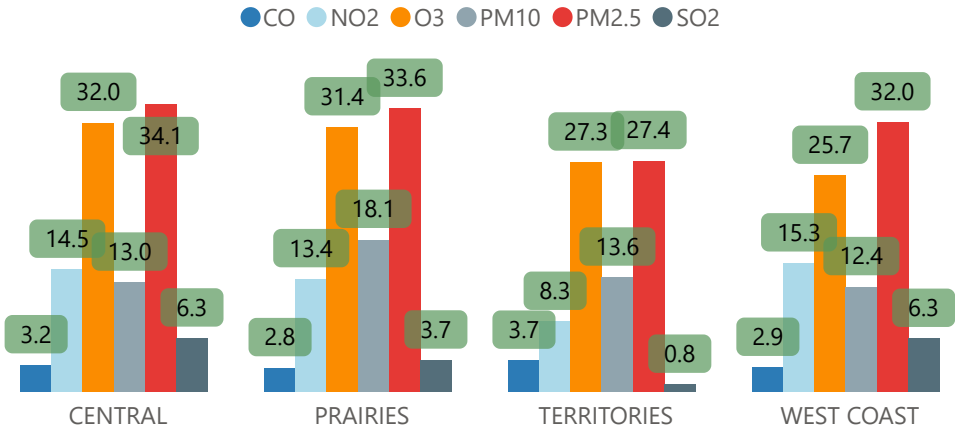
2024

Is Atlantic

ATLANTIC

REST OF CAN...

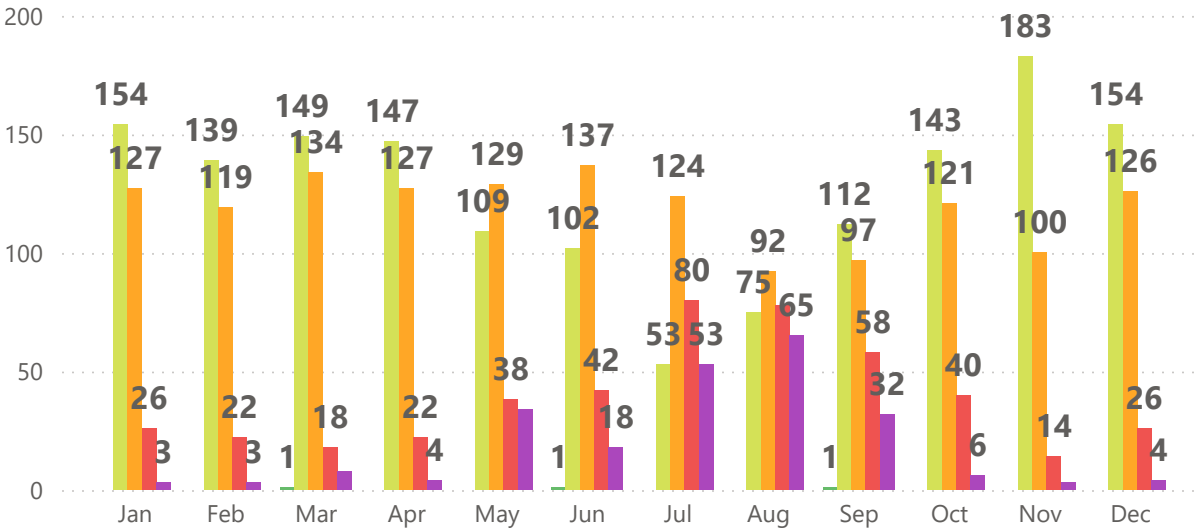
Avg Pollutant AQI



Worst Day: Pollutant AQI

Province_Code	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
⊕ AB	172	194	283	208	300	296	297	297	300	272	233	178
⊕ BC	215	197	254	193	288	210	280	295	297	283	189	242
⊕ NT	79	105	156	167	277	214	295	297	277	157	70	94
⊕ MB	197	255	254	213	268	245	289	259	288	185	205	171
⊕ SK	116	147	126	163	286	284	279	239	279	224	139	114
⊕ ON	106	124	173	128	175	279	166	169	135	163	138	137
⊕ QC	157	239	139	121	150	273	159	152	140	153	156	188
⊕ NU	260	260	79	55	61	44	54	260	62	88	79	103
⊕ YT	136	113	81	64	71	105	151	90	59	115	80	122

Days in AQI Category



Pollutant

- co
- no2
- o3
- pm10
- pm2.5
- so2

Station_Type

AQI St...